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November 13, 2025

Attn: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

10 CFR 50.73

**SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-387/2025-001-01
UNIT 1 LICENSE NO. NPF-14
PLA-8198**

Docket No. 50-387

Attached is Licensee Event Report (LER) 50-387/2025-001-01. The LER supplements the earlier report of an event involving a manual scram following the partial loss of Main Transformer cooling that was determined to be reportable in accordance with 10 CFR 50.72(b)(2)(iv)(A), 10 CFR 50.72(b)(2)(iv)(B) and 10 CFR 50.72(b)(3)(iv)(A). This event was also reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) as an event that resulted in a manual actuation of the Reactor Protection System (including a reactor scram) and actuation of an emergency core cooling system.

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new or revised regulatory commitments.

A handwritten signature in black ink, appearing to read "E. Casulli".

E. Casulli

Attachment: LER 50-387/2025-001-01

Copy: NRC Region I
Mr. R. Wehrmann, NRC Senior Resident Inspector
Mr. R. Guzman, NRC Project Manager
Mr. M. Shields, PA DEP/BRP

Attachment to PLA-8198

LER 50-387/2025-001-01

NRC FORM 366

(04-02-2024)

U.S. NUCLEAR REGULATORY COMMISSION

APPROVED BY OMB: NO. 3150-0104

EXPIRES: 04/30/2027

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Susquehanna Steam Electric Station (SSES) Unit 1

☒ 050☐ 052**2. Docket Number**

00387

3. Page

1 OF 3

4. Title

Manual Reactor Scram Following Loss of Cooling to Main Transformer

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
07	22	2025	2025	- 001 -	01	11	13	2025	Facility Name	☐ 050
									Facility Name	☐ 052

9. Operating Mode

1

10. Power Level

96

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	20.2203(a)(2)(vi)	10 CFR Part 50	50.73(a)(2)(ii)(A)	50.73(a)(2)(viii)(A)	73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).**12. Licensee Contact for this LER****Licensee Contact**

Andrew Sabisch, Nuclear Regulatory Affairs

Phone Number (Include area code)

570-542-1461

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)**15. Expected Submission Date**

Month

Day

Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On July 22, 2025, at approximately 15:10, Susquehanna Steam Electric Station (SSES) Unit 1 was manually scrambled following a partial loss of cooling to the main transformer (one of two cooling fan groups were found tripped). The event was reported by Event Notification 57822 in accordance with 10 CFR 50.72(b)(2)(iv)(A), 10 CFR 50.72(b)(2)(iv)(B) and 10 CFR 50.72(b)(3)(iv)(A). This event is also reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) as an event that resulted in a manual actuation of the Reactor Protection System (including a reactor scram) and actuation of the emergency core cooling system. Following restoration of both cooling fan groups, SSES Unit 1 was restored to Mode 1 operation on July 25, 2025. The root cause of the event was determined to be that Operations leadership did not ensure the Operators were prepared to respond to a loss of cooling event on the main transformer to minimize the potential for a required manual scram. There were no actual consequences to the health and safety of the public as a result of this event.



**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 60 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME		☒ 050	2. DOCKET NUMBER	3. LER NUMBER						
Susquehanna Steam Electric Station (SSES) Unit 1		☐ 052	00387	<table border="1"><tr><td>YEAR</td><td>SEQUENTIAL NUMBER</td><td>REV NO.</td></tr><tr><td>2025</td><td>001</td><td>01</td></tr></table>	YEAR	SEQUENTIAL NUMBER	REV NO.	2025	001	01
YEAR	SEQUENTIAL NUMBER	REV NO.								
2025	001	01								

NARRATIVE

CONDITIONS PRIOR TO THE EVENT

Unit 1: Mode 1, approximately 96% Rated Thermal Power
Unit 2: Mode 1, approximately 100% Rated Thermal Power

There were no inoperable structures, systems, or components at the start of the event that contributed to the event

EVENT DESCRIPTION

On July 22, 2025, a Main Transformer 1X101 [EIS System Code/ Component Code - EL/ XFMR] trouble alarm was received in the Control Room and Operators entered a 10-minute manual scram requirement in accordance with the Alarm Response procedure.

The non-licensed operator (NLO) responding in the field to the Main Transformer correctly determined that only a partial loss of cooling had occurred with Group 1 cooling fans [IEEE Component Code - FAN] not running and Group 2 cooling fans in service. However, imprecise communications resulted in operators in the Control Room and Work Control Center (WCC) incorrectly believing that a total loss of cooling had occurred. As a result, a second NLO, dispatched to the transformer by the WCC, attempted to restore transformer cooling. Failing to recognize that partial cooling for the transformer was in fact in service, the second NLO secured the running Group 2 cooling fans while attempting to switch to the alternate power supply resulting in the total loss of transformer cooling.

At approximately 15:10, control room operators manually scrambled SSES Unit 1 following a loss of cooling to the Main Transformer.

All systems responded normally following the manual scram. High Pressure Coolant Injection [EIS System Code - BJ] actuated briefly to assist in restoring reactor water level. The 1B Reactor Feedwater Pump [EIS System Code/ Component Code - SJ/ P] stabilized and maintained reactor water level. Decay heat was removed by discharging steam to the main condenser using the turbine bypass valves.

The event was reported by Event Notification 57822 in accordance with 10 CFR 50.72(b)(2)(iv)(A), 10 CFR 50.72(b)(2)(iv)(B) and 10 CFR 50.72(b)(3)(iv)(A). This event is also reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) as an event that resulted in a manual actuation of the Reactor Protection System (including a reactor scram) and actuation of the emergency core cooling system.

Following restoration of both cooling fan groups, SSES Unit 1 was returned to Mode 1 operation on July 25, 2025.

CAUSE OF THE EVENT

The direct cause of the event was ineffective communication and direction between control room and field personnel responding to a partial loss of main transformer cooling (Group 1 Cooling). This resulted in incorrect operator actions being taken that caused a complete loss of transformer cooling.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME Susquehanna Steam Electric Station (SSES) Unit 1	☒ 050	2. DOCKET NUMBER 00387	3. LER NUMBER		
	☐ 052		YEAR 2025	SEQUENTIAL NUMBER 001	REV NO. 01

NARRATIVE**CAUSE OF THE EVENT cont.**

The root cause of the event was determined to be that Operations leadership did not ensure the Operators were prepared to respond to a loss of cooling event on the main transformers to minimize the potential for a required manual scram. This included not ensuring that procedures, training, and readiness were aligned to restore cooling in a total loss of cooling event and to protect the in-service cooling group in a partial loss of cooling event

ANALYSIS / SAFETY SIGNIFICANCE

The actual consequence of this event was a manual reactor scram. All cooling to Main Transformer 1X101 was actually lost for approximately six minutes. No damage occurred to the transformer as confirmed by a review of data from the installed Dissolved Gas Analyzer and transformer winding temperatures.

Plant response to the scram was proper and per design. There were no Diesel Generator starts nor Safety Relief Valve actuations during the event. Fuel integrity was not challenged during the transient as evidenced by Main Steam Line and Offgas radiation levels lowering as expected following the scram. In-plant radiation levels and vent release rates were all normal based on plant conditions and within acceptable limits.

The scram was reviewed with regard to Final Safety Analysis Report Chapter 15 and determined to be fully bounded by the analysis. Section 15.2.3 specifically addresses main turbine trip with main turbine bypass operable and classifies this as a non-limiting event. The Main Turbine Bypass System operated as designed during this event to control reactor pressure. All safety systems were observed to operate as designed

There were no equipment consequences due to this event and no impact to the safety and health of the public. This event will not be counted as a Safety System Functional Failure under the NRC Reactor Oversight Process Performance Indicator program.

CORRECTIVE ACTIONS

Key corrective actions taken in response to the event included repairing the breaker that caused the loss of Main Transformer 1X101 Group 1 cooling fans, conducting stand-downs with all of the operating crews to communicate lessons learned and reinforce Operations standards. Additionally, formal training on the use of 3-part communication will be completed followed by observations of Operations personnel in the field and control room focused on 3-part communication, teamwork and resource management. Lastly creation of a new Off-Normal procedure to address partial or total loss of transformer cooling will be completed.

COMPONENT FAILURE INFORMATION

NONE

PREVIOUS OCCURRENCES

NONE